Appendix_Exercises.ipynb > #[STUDENT CONFIG] Jupyter Notebook

Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ... .venv (3.14.7) (Python 3.14.7)

```python
def merge(a, b):
    out = []
    while i < len(a) and j < len(b):
        comps += 1
        if a[i] <= b[j]:
            out.append(a[i]); i += 1
        else:
            out.append(b[j]); j += 1
    out.extend(a[i:]); out.extend(b[j:])
    return out, comps

A = [SEED_DIGIT + ID_SUM * i for i in range(1, 9)]
B = [ARTIST_LENGTH + NAME_LENGTH * i for i in range(1, ARTIST_LENGTH + 1)]
merged, comps = merge(A, B)
print("Dept A:", A)
print("Dept B:", B)
print("Consolidated:", merged)
print("Records before:", len(A), "+", len(B), "=", len(A) + len(B), "| after:", len(merged))
print("Comparisons:", comps)
print("Ordering verified:", merged == sorted(A + B))
```

Click to add a breakpoint 0

[7] Python

```
Dept A: [31, 55, 79, 103, 127, 151, 175, 199]
Dept B: [16, 22, 28, 34, 40, 46, 52, 58, 64, 70]
Consolidated: [16, 22, 28, 31, 34, 40, 46, 52, 55, 58, 64, 70, 79, 103, 127, 151, 175, 199]
Records before: 8 + 10 = 18 | after: 18
Comparisons: 12
Ordering verified: True
```